Gradient-intercept form

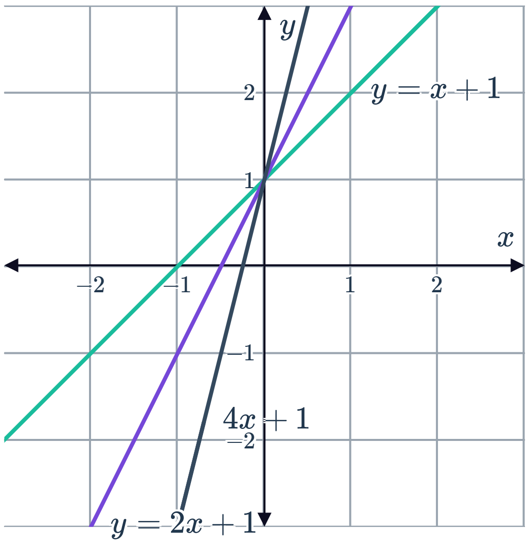
1 Consider the equation of a line $y = mx + c$.

a What does m represent?

b What does c represent?

2 Consider the following three linear equations with their graphs plotted on a number plane:

- Equation 1: $y = x + 1$
- Equation 2: $y = 2x + 1$
- Equation 3: $y = 4x + 1$



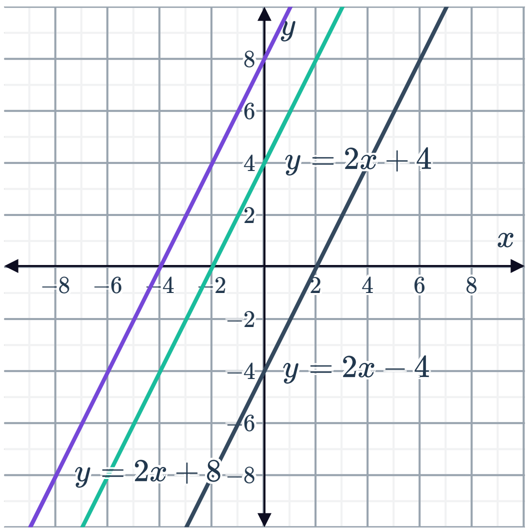
a What do all of the equations have in common?

b What do all of the graphs have in common?

c Describe all lines that have the form: $y = mx + 1$

3 Consider the following three linear equations with their graphs plotted on a number plane:

- Equation 1: $y = 2x + 4$
- Equation 2: $y = 2x + 8$
- Equation 3: $y = 2x - 4$



a What do all of the equations have in common?

b What do all of the graphs have in common.

c Describe all lines that have the form: $y = 2x + c$

4 Determine whether or not the gradients of the following pairs of equations are equal:

a
i $y = 7x$
ii $y = 7x - 4$

b
i $y = 7x + 2$
ii $y = \frac{x}{7} + 4$

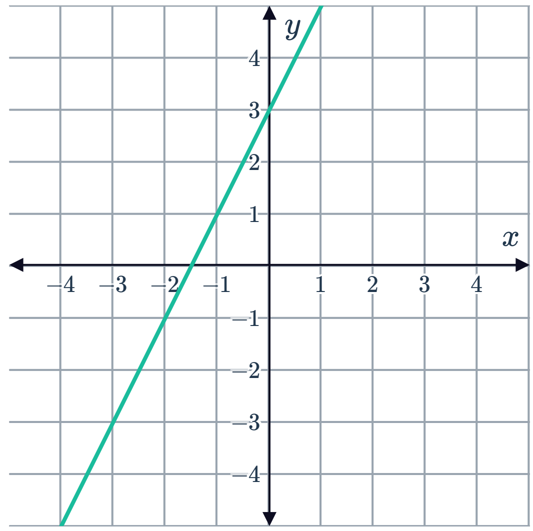
c
i $y = 6 - 7x$
ii $y = 6x + 7$

d
i $y = 4 + 7x$
ii $y = 7x - 10$

5 Consider the line plotted below:

a State the the y-intercept.

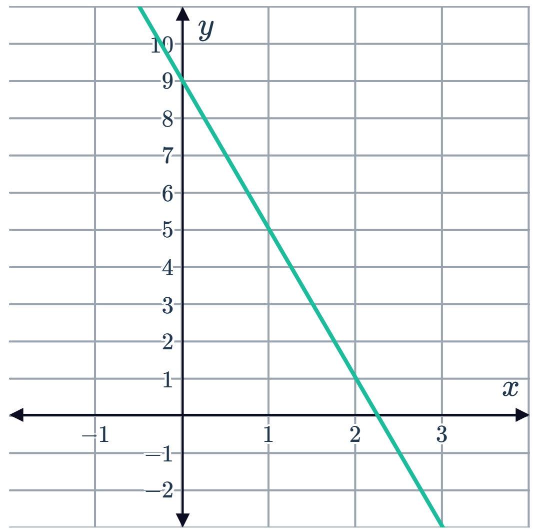
b State the gradient.



6 Consider the line plotted below:

a By how much does the y-value change as the x-value increases by 1?

b State the gradient of the line.



7 State whether or not the following lines have a y-intercept:

a $y = 2$ b $5x = 4y$ c $y = -4x + 1$ d $x = 1$

e $y = 6x - 2$ f $y = 3x$ g $2x + 6y = 3$ h $y + 4 = -5x$

8 Consider the following linear equations:

i State the value of the gradient, m. ii State the y-intecept, c.

a $y = -2x + 9$ b $y = -4x - 8$ c $y = 8x + 6$ d $y = -1 - \dfrac{9x}{2}$

e $3y = 12x - 15$ f $-9x + 9y = 27$ g $3x - 10y = -2$ h $16x + 12y = 10$

9 Sketch the following lines on a number pl

a The line with a y-intercept of -2 and gradient of -3.

b The line with a y-intercept of 3 and gradient of $-\frac{3}{2}$.

c The line $y = 2x + 5$.

d The line $y = \frac{1}{2}x - 1$.

10 For each of the following equations:

i Find the y-value of the y-intercept of the line.

ii Find the x-value of the x-intercept of the line.

iii Find the value of y when $x = 3$.

iv Sketch the line on a number plane.

a $y = -2x$

b $y = \frac{2x}{3} - 4$

11 For each of the following lines:

i Find the y-coordinate of the y-intercept of the line.

ii Hence, write the equation of the line in gradient-intercept form.

iii Find the x-coordinate of the x-intercept of the line.

iv Sketch the line on a number plane.

a A line has gradient $\frac{4}{5}$ and passes through the point $(-10, 4)$.

b A line has gradient -2 and passes through the point $(3, -8)$.

Equation of a line

12 A line has a gradient of -3 and intercepts the y-axis at -2.

a Find the equation of the line in the form $y = mx + c$.

b State whether the point $(-2, 4)$ lies on this line.

13 A line has a gradient of -3 and cuts the y-axis at 8.

a Find the equation of the line in the form $y = mx + c$.

b State whether the point $(8, -31)$ lies on this line.

14 Find the equations of the following in the [redacted] $y = mx + c$:

a A line that has the same gradient as $y = 9 - 8x$ and the same y-intercept as $y = -5x - 3$.

b A line whose gradient is 2 and crosses the y-axis at 5.

c A line whose gradient is -6 and crosses the y-axis at 9.

d A line whose gradient is -8 and crosses the y-axis at -9.

e A line whose gradient is $-\dfrac{3}{4}$ and intercepts the y-axis at 3.

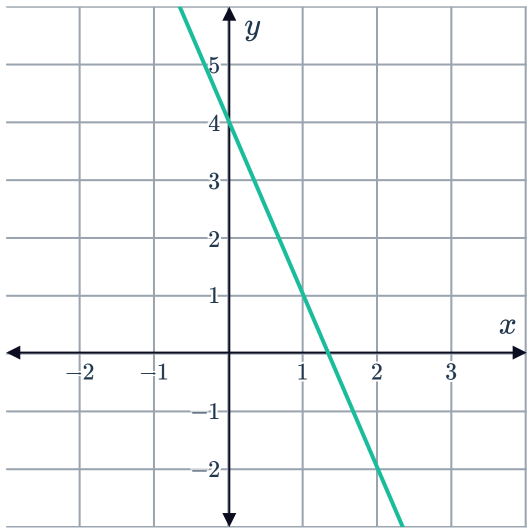
f A line whose gradient is $\dfrac{4}{3}$ and goes through the point $(0, 3)$.

g A line that has a gradient of -2 and passes through $(-6, -3)$

h A line whose gradient is 8 and goes through the point $(0, -4)$.

i A line whose gradient is 0 and goes through the point $\left(0, \dfrac{2}{13}\right)$.

15 Consider the line plotted on the number plane.



a State the values of:

i The gradient, m.

ii The y-intercept, c.

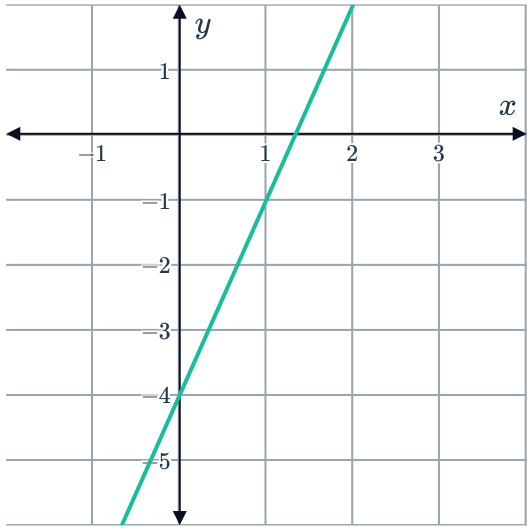
b Find the equation of the line in gradient-intercept form.

c Find the value of y when $x = 27$.

16 Consider the line plotted on the number p

a Find the equation of the line in gradient-intercept form.

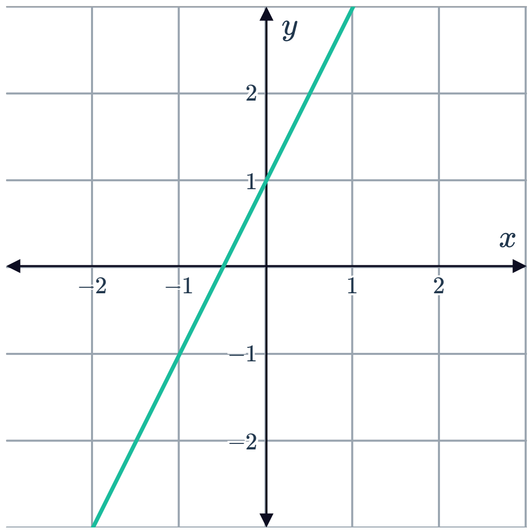
b Find the value of y when $x = 50$.



17 Consider the line plotted on the number plane.

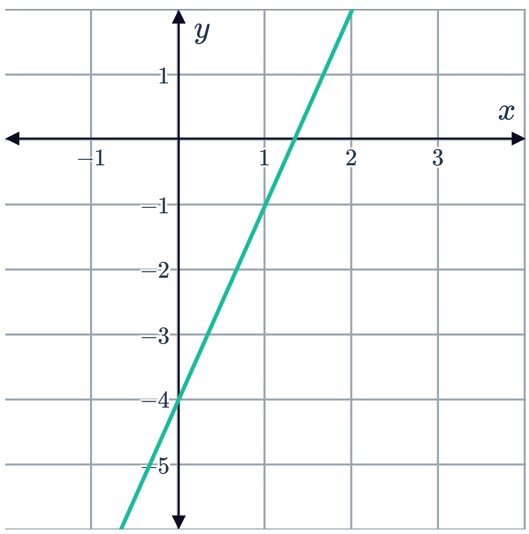
a Find the equation of the line in gradient-intercept form.

b Find the value of y when $x = 29$.

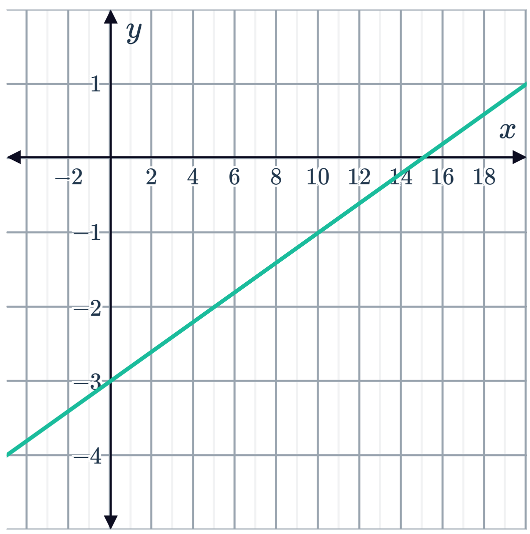


18 Find the equations of the following lines in gradient-intercept form:

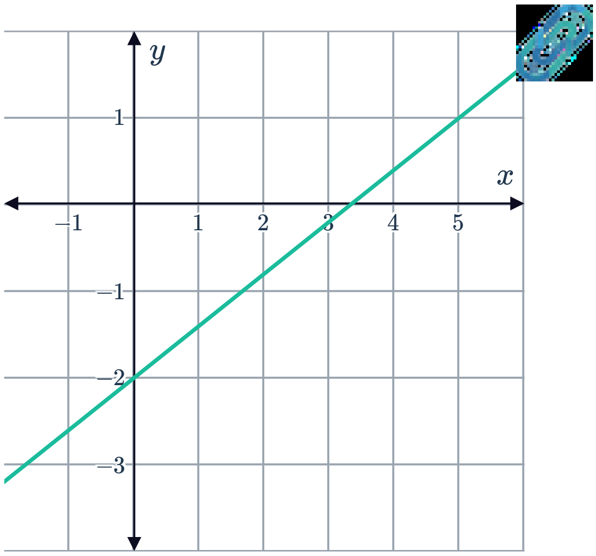
a



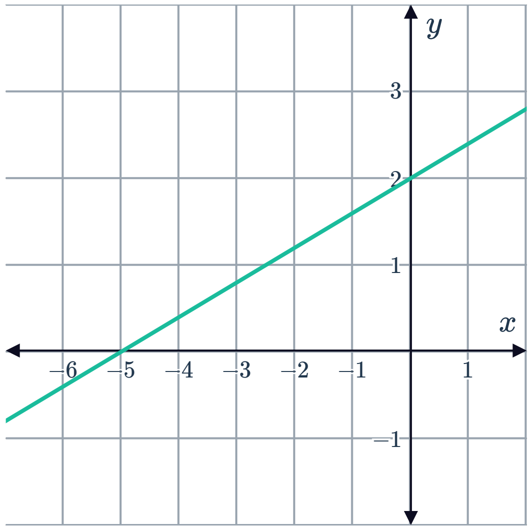
b



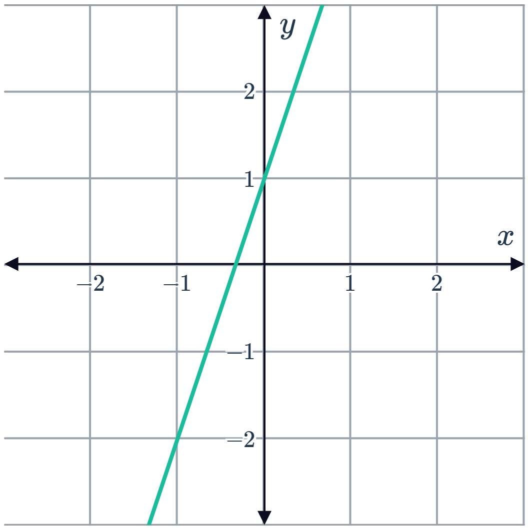
c


y
x
1
−1
−2
−3
−1
1
2
3
4
5

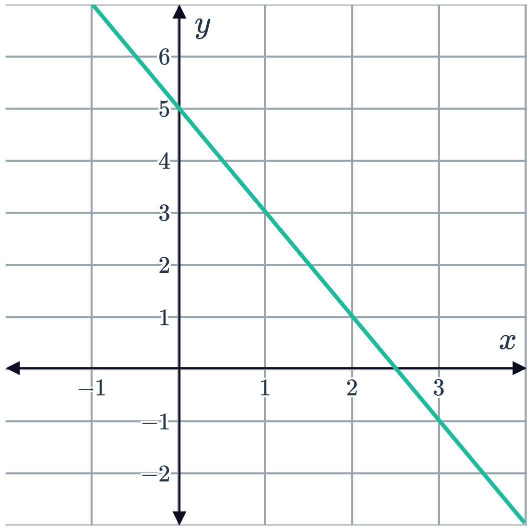
d


y
x
3
2
1
−1
−6
−5
−4
−3
−2
−1
1

e


y
x
2
1
−1
−2
−2
−1
1
2

f


y
x
6
5
4
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2
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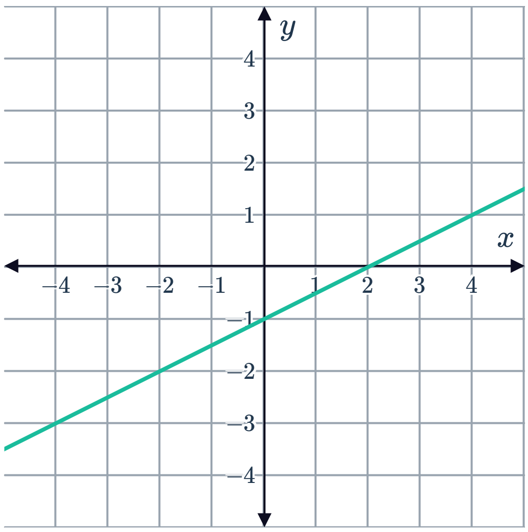
−1
−2
−1
1
2
3

g


y
x
4
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1
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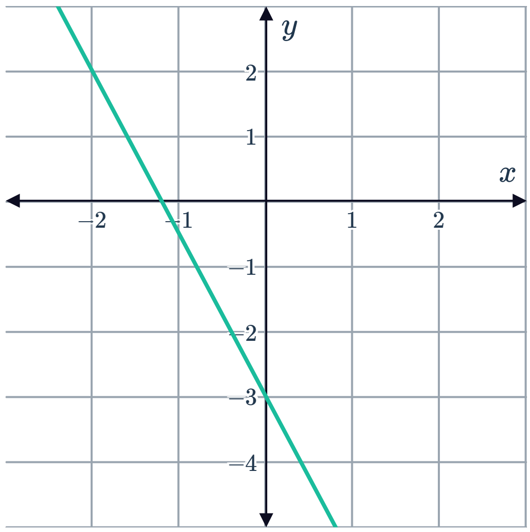
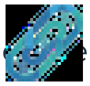
4

h


y
x
2
1
−1
−2
−3
−4
−2
−1
1
2

19 For the following lines passing through thn two points:

i Hence, state the gradient of the line.

ii Find the equation of the line in the form $y = mx + c$.

a $(0, 2)$ and $(2, 6)$

b $(0, -9)$ and $(5, 1)$

c $(0, 2)$ and $(-7, 44)$

d $(0, -3)$ and $(1, 4)$

20 For each of the following linear equations:

i Rewrite it in the form $y = mx + c$.

ii State the gradient of the line, m.

iii State the y-intercept of the line, c.

a $y = \dfrac{-4x + 16}{4}$

b $9x - y - 8 = 0.$

c $y = 3\,(4x - 3)$

d $y = 6\,(3x - 2)$

e $3x - 9y - 27 = 0.$

f $3x - 4y - 28 = 0$

21 Consider the lines with the following equations:

- Line A: $5x + 3y + 5 = 0$
- Line B: $7x + 6y - 3 = 0$

a Express the lines in the form $y = mx + c$.

b State which line is steeper, A or B.

22 Determine which of the following lines are steeper: $2x + 5y - 5 = 0$ or $4x + 4y + 1 = 0$.

23 A straight line has gradient -1 and goes through the points $(0, 2)$ and $(a, -6)$.

a Write the equation of the line in the form $y = mx + b$.

b Find the value of a.

Gradients of parallel lines

24 Determine whether the following statements about two parallel lines are true or false.

a The y-value is changing at the same rate on both lines.

b They intersect at one point.

c They have the same value of c in $y = mx + c$.

d They have the same value of m in $y = mx + c$.

e They are equidistant from each other.

25 State whether the given pairs of lines are parallel:

a $y = -2x - 5$ and $y = -2x - 8$

b $y = 7x + 8$ and $y = -5x + 8$

c $y = -3x - 2$ and $y = -3x + 9$

d $y = -6x - 5$ and $y = -6x$

26 Find the gradient of the following lines:

a A line parallel to a line with gradient -2.

b Any line that is parallel to $y = -7 + 4x$.

27 State whether the following lines are parallel to $y = 7x + 3$.

a $y = 7x - 3$

b $y = 6x + 3$

c $y = 7x$

d $y = -7x + 3$

28 State whether the following lines are parallel to $y = -3x + 2$.

a $y = 3x$

b $y = -\dfrac{2x}{3} + 8$

c $-3y - x = 5$

d $y = -10 - 3x$

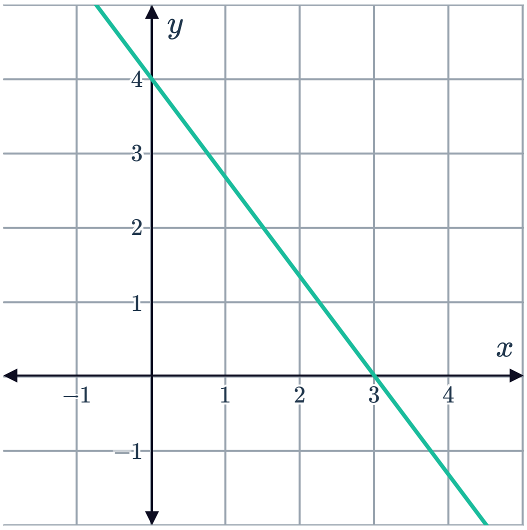
Point-gradient formula

29 For the following graphs:

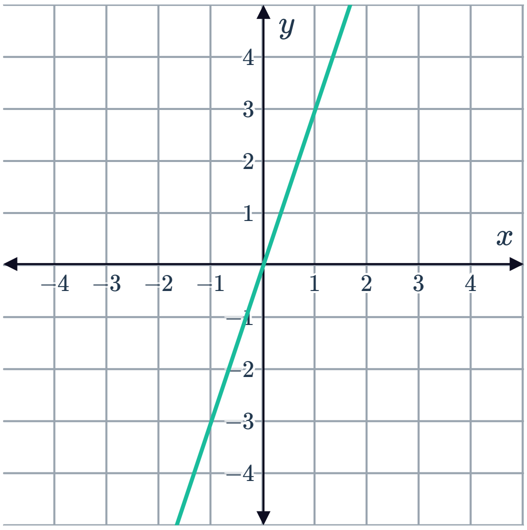
i State the value of the x-intercept.

ii State the value of the y-intercept.

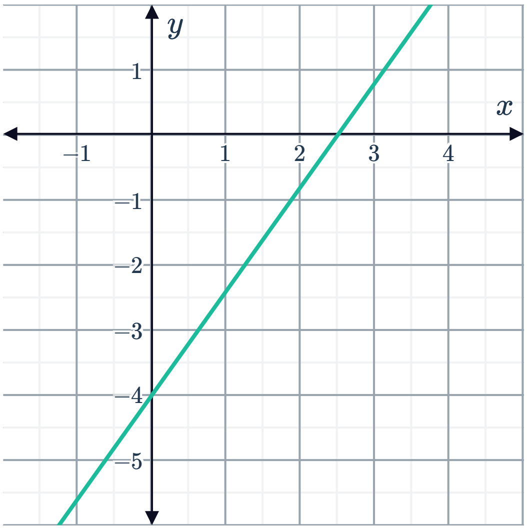
a



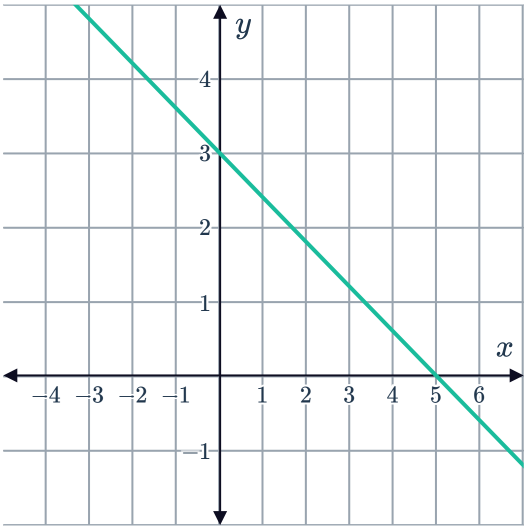
b



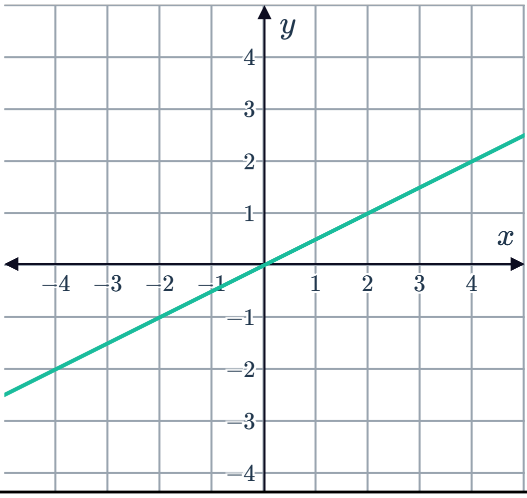
c



d



e



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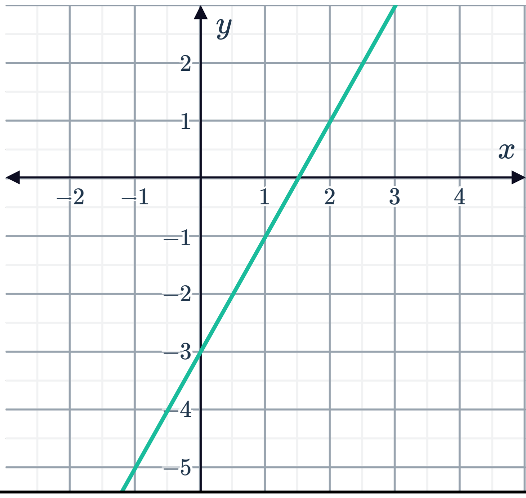

30 For each of the following tables of values:

i Find the gradient, m.

ii Find the y-intercept, c.

iii Write the equation of the line expressing the relationship between x and y.

iv Complete the table of values.

a

x	0	1	2	3	4	24
y	0	4	8	12	16	

b

x	0	1	2	3	4	21
y	9	14	19	24	29	

c

x	0	1	2	3	4	25
y	-23	-21	-19	-17	-15	

d

x	0	1	2	3	4	70
y	27	22	17	12	7	

31 For each of the following tables of values:

i Find the equation of the line expressing the relationship between x and y.

ii Complete the table of values.

a

x	1	2	3	4	19
y	5	10	15	20	

b

x	1	2	3	4	16
y	-3	-6	-9	-6	

c

x	1	2	3	4	60
y	1	4	7	10	

d

x	1	2	3	4	80
y	-14	-22	-30	-38	

e

x	1	2	3	4	-15
y	82	74	66	58	

32 Find the equation that corresponds to each of the following tables:

a

x	-1	0	1	2	3
y	7	4	1	-2	-5

b

x	1	2	3	4	5
y	5	8	11	14	17

c

x	-8	-7	-6	-5	-4
y	-37	-32	-27	-22	-17

d

x	3	4	5	6	7
y	-1	-3	-5	-7	-9

33 Find the equation of the following lines:

a A line that passes through the point A $(-5,-4)$ and has a gradient of -4.

b A line that passes through the point A $\left(-\frac{4}{5},-4\right)$ and has a gradient of 2.

c A line that passes through Point $A\,(-4,3)$ and has a gradient of 4.

d A line that passes through Point $A\,(7,-6)$ and has a gradient of -3.

e A line that passes through the point A $(3,5)$ and has a gradient of $-\frac{5}{2}$.

f A line that passes through the point A $(4,3)$ and has a gradient of $-3\frac{1}{3}$.

g A line that passes through the point $A\,(-4,3)$ and has a gradient of -9.

h A line that passes through the point $A\left(-\frac{5}{9},7\right)$ and has a gradient of 7.

i A line that passes through the point $A\,(8,1)$, and has a gradient of $\frac{5}{2}$.

j A line that passes through the point $A\,(-4,5)$ and has a gradient of $3\frac{1}{2}$.

34 For each of the following lines:

i Find the equation of the line.

ii Sketch the graph of the line.

a A line has gradient 2 and passes through the point $(-5,-3)$.

b A line has gradient $-\frac{3}{2}$ and passes through the point $(-2,\ 2)$.

c A line has gradient $-\frac{2}{5}$ and passes through the point $(-10,2)$.

d A line has gradient -3 and passes through the point $(2,-12)$.

35 Consider the line with equation $2x+y-8=0$.

a Find the x-intercept of the line.

b Hence, find the equation of a line with a gradient of -4 that passes through the x-intercept of the given line.

36 Find, in general form, the equation of a line which has a gradient of $\frac{4}{7}$ and cuts the x-axis at -10.

37 For each of the following lines:

i Find the gradient of the line.

ii Find the equation of the line.

a A line passes through the points $(2, -7)$ and $(-5, 6)$.

b A line passes through the points $(3, -3)$ and $(5, -11)$.

c A line passes through the points $A\,(-6, 7)$ and $B\,(-8, -4)$.

38 Identify which of the following equations of straight lines have a gradient of 5 and pass through the point $A\,(-1, -4)$:

a $\dfrac{y+4}{x+1} = 5$

b $\dfrac{x+1}{y+4} = 5$

c $\dfrac{-4-y}{-1-x} = 5$

d $\dfrac{y+1}{x+4} = 5$

39 Write down the equations of three lines that pass through the point $(1, 3)$. Explain how your lines are different.

Applications

40 A line has a gradient of $\dfrac{3}{10}$ and passes through the midpoint of $A\,(-6, -6)$ and $B\,(8, 8)$.

a Find the coordinates of M, the midpoint of AB.

b Find the equation of the line in general form.

41 Consider the lines L_1, $y = -4x + 5$, and L_2, $y = x - 1$.

a Find the midpoint M of their y-intercepts.

b Find the equation of the line that goes through the point M and has gradient $\dfrac{1}{3}$. Express the equation in general form.

42 A circle with centre $C\,(11, 13)$ has a diameter with end points $A\,(5, 14)$ and $B\,(p, q)$.

a Find the value of p.

b Find the value of q.

c Find the equation of the line passing through B with gradient $\dfrac{9}{2}$.